$\tau \to \mu\gamma$ and $\tau \to e\gamma$ in the S_3 3HDM: allowed at the exact vacuum, and no new constraint in the soft one

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Abstract

This report completes the three radiative LFV decays $\ell_i \to \ell_j\gamma$ on the same 2028 soft-broken points as the $\mu \to e\gamma$ report, and adds the 1973 exact-S_3 points, which $\mu \to e\gamma$ cannot test. The diagrams are the same; only the flavour indices and the τ normalization change, and both are checked against independent sources. At the exact vacuum the residual $\mathbb{Z}_2$ splits the two channels: $\tau \to e\gamma$ vanishes identically, while $\tau \to \mu\gamma$ is allowed and generic. The Belle limit [1] excludes only 0.72% of the exact configurations, far fewer than $h \to \tau\mu$, but *not a subset* of them: where a non-SM $\mathbb{Z}_2$-even scalar is light, $\tau \to \mu\gamma$ excludes what the Higgs decay allows. In the soft vacuum the τ modes add nothing beyond MEG II: every scalar point keeps a τ-allowed configuration, and the counts stay at **618 of 2028** (606 with the Higgs LFV limits). Where every limit holds the τ rates stay below 5×10^{-10}, an order of magnitude under the projected reach of Belle II.

1 Summary

Everything below comes from `research/thdm_s3/08_tau_to_l_gamma.ipynb`, with the numerics in `radiative.py` and `soft_decays.py`; the two scans run in `compute_tau_l_gamma.py`.

1. **The generalization to $\ell_i \to \ell_j\gamma$ is checked**, not assumed (§2).
2. **For an SM-like Higgs the Higgs decay wins in the τ sector**, by about 12× in the coupling: the reverse of the $e\mu$ sector (§3).
3. **Exact S_3: $\tau \to e\gamma = 0$, $\tau \to \mu\gamma \neq 0$**, and Belle removes a light-scalar corner that $h \to \tau\mu$ misses (§4).
4. **Soft vacuum: no point removed beyond MEG II** (§5).
5. **A prediction for Belle II** (§6).

2 From $\mu \to e\gamma$ to $\ell_i \to \ell_j\gamma$

The amplitude of the previous report carries over with the decaying lepton i and the final one j as arguments. The neutral one loop becomes

$$c_L = \sum_f \frac{\frac{1}{2} I_1(t_f)\,(m_j Y_{jf} Y_{if}^* + m_i Y_{fj}^* Y_{fi}) + I_2(t_f)\, m_f Y_{fj}^* Y_{if}^*}{4 m_i m_\phi^2}, \qquad A = \frac{c}{4\pi^2},$$

the charged loop is $A_L = -(Y_C^\dagger Y_C)_{ji}/(192\pi^2 m_{H^\pm}^2)$, and the Barr–Zee terms take (Y_{ij}, Y_{ji}) and $1/m_i$ in place of $(Y_{\mu e}, Y_{e\mu})$ and $1/m_\mu$. Two things are new physics rather than plumbing:

- the internal-τ chirality flip, which dominates the one loop of $\mu \to e\gamma$ ($\propto m_\tau/m_\mu$), is **not enhanced** for a decaying τ (m_τ/m_τ, times a logarithm);
- the width is normalized to $\Gamma(\tau \to \ell\nu\bar{\nu})$ and multiplied by its branching ratio [4], Eq. (17):

$$\mathcal{B}(\tau \to \ell\gamma) = \frac{48\pi^3\alpha\,(|A_L|^2 + |A_R|^2)}{G_F^2}\,\mathcal{B}(\tau \to \ell\nu\bar{\nu}).$$

check	against	agreement
$(i,j) = (\mu, e)$, 25 points	stored $\mu \to e\gamma$ results (4 figures)	3.5×10^{-4}
relabelling $(e,\mu,\tau) \to (\mu,\tau,e)$	the (μ, e) code	3×10^{-16}
closed-form one loop, $\tau \to \mu\gamma$	[5] (A3), external masses kept	3×10^{-6} / 2×10^{-3} (700 / 25 GeV)
one loop, internal τ only	[4] Eq. (19)	10^{-4}
τ normalization	[5]'s lifetime normalization, [9]	0.2%
$\tau \to \mu\gamma$, $\tau \to e\gamma$ bounds	[5]: 0.016, 0.014	0.0159, 0.0136

The relabelling test uses the couplings of a real soft point and shows that no fixed e or μ index survived in the general code. The [4] Eq. (19) check also confirms where m_i and m_f sit: the flip and non-flip terms there carry the same mass, and a version with the m_τ/m_μ flip factor of $\mu \to e\gamma$ carried over fails it. The [5] bounds are reproduced with their W-loop number (the previous report found ours 14% above it). In the τ channels the one loop is comparable to the two loop, so this oracle also fixes their relative sign; flipping it fails the comparison.

3 An SM-like Higgs: the Higgs decay is stronger here

At the current limits and pole m_t, an SM-like Higgs needs

$$\sqrt{|Y_{\tau\mu}|^2 + |Y_{\mu\tau}|^2} < 0.013 \text{ (Belle)}, \qquad \sqrt{|Y_{\tau e}|^2 + |Y_{e\tau}|^2} < 0.011 \text{ (BaBar)},$$

while the CMS searches for $h \to \tau\mu$ and $h \to \tau e$ [7] need 0.0011 and 0.0013. In the τ sector **the Higgs decay is** $12\times$ **and** $8\times$ **stronger in the coupling**. This is the reverse of the $e\mu$ case, where MEG II bounds the coupling at 7×10^{-7} against 10^{-4} for $h \to e\mu$. The τ modes can still matter where the Higgs decay is blind: the LFV couplings of the *non-SM* scalars.

4 The exact-S_3 vacuum: $\tau \to e\gamma$ vanishes, $\tau \to \mu\gamma$ does not

The previous report traced the zero of $\mu \to e\gamma$ at the exact vacuum to its residual $\mathbb{Z}_2$. Under it the electron and the in-plane non-vacuum scalars (h_0, one A, one $H^\pm$) are odd, and μ, τ even. So $\tau \to e\gamma$, with one odd lepton, is forbidden, while $\tau \to \mu\gamma$, with none, is allowed through the even scalars. The exact points, which $\mu \to e\gamma$ cannot test, meet their first radiative constraint here.

Each of the 1973 exact points is evaluated at $\varphi = \pi/3$ with every free soft term zero, on the draft lepton branch, at 60 values of $\mu_3 \in (m_\mu, m_\tau)$: 118 380 evaluations.

- $\mathcal{B}(\tau \to e\gamma) < 10^{-33}$ and $\mathcal{B}(\mu \to e\gamma) < 10^{-29}$ everywhere, which is round-off, although single $e\tau$ couplings reach 0.09.

- Every evaluation shows the pattern behind it: three odd states with no $\tau\mu$ coupling, and four even states with no coupling to the electron.
- Off the alignment, with one exact point and the free soft terms still zero, $\mathcal{B}(\tau \to e\gamma) \propto (60^\circ - \varphi)^2$ (fitted slope 2.00), as for $\mu \to e\gamma$.
- $\tau \to \mu\gamma$ is generic: median 2.1×10^{-11}, maximum 1.1×10^{-4}.

	exact-vacuum evaluations excluded
Belle, $\tau \to \mu\gamma$ [1]	0.72%
CMS, $h_{\rm SM} \to \tau\mu$ [7]	18.5%
Belle but not CMS	460 evaluations, at 20 points

The light-scalar corner. The Belle exclusion is thin but not contained in the CMS one. In all 460 evaluations that only Belle excludes, the one loop is the largest piece, and a non-SM neutral $\mathbb{Z}_2$-even scalar is light: 6.7–223 GeV, against a median of 302 GeV over all exact points. Its $\tau\mu$ coupling enters with $1/m_\phi^2$, and the SM-like Higgs decay cannot see that state. The closed-form one loop still holds there, to 1.3×10^{-2} at 6.7 GeV against the full [5] integral. No exact point loses its whole μ_3 range.

The S_3 image. $\varphi \to 2\pi/3 - \varphi$, with both soft doublets reflected, is the same physics: all three branching ratios agree at a point and at its image to 4×10^{-10} (20 points, every configuration).

5 The soft-broken scan

Every point × the 73-value μ_3 grid × every lepton branch gives 417 992 configurations, with all three channels and the SM-like state's $h \to \ell\ell'$ rates computed together. The medians are $\mathcal{B}(\tau \to \mu\gamma) = 2.2 \times 10^{-12}$ and $\mathcal{B}(\tau \to e\gamma) = 2.3 \times 10^{-12}$, and only 0.020% and 0.029% of the configurations exceed Belle and BaBar [2].

limits applied	scalar points with an allowed configuration
$\tau \to \mu\gamma$ and $\tau \to e\gamma$	2028 / 2028
MEG II [6]	618 / 2028
all three radiative	**618 / 2028**
all three radiative and the $h \to \ell\ell'$ limits [7, 8]	606 / 2028

Adding both τ limits to MEG II removes no point. Configuration by configuration, measured against its own limit, $\tau \to \mu\gamma$ is a median 9×10^{-3} as restrictive as $h \to \tau\mu$, and only 108 configurations are excluded by a τ mode while allowed by every Higgs LFV limit. The W-loop Barr–Zee diagram is the largest piece of both τ channels in 93% of configurations, as for $\mu \to e\gamma$.

Robustness. The same three variants as in the $\mu \to e\gamma$ report, built from the same amplitudes for every channel:

variant	scalar points allowed by all three radiative limits
baseline	618
no quark Barr–Zee	504
quark couplings with opposite sign	453
W loop scaled to [5]'s 0.11	684

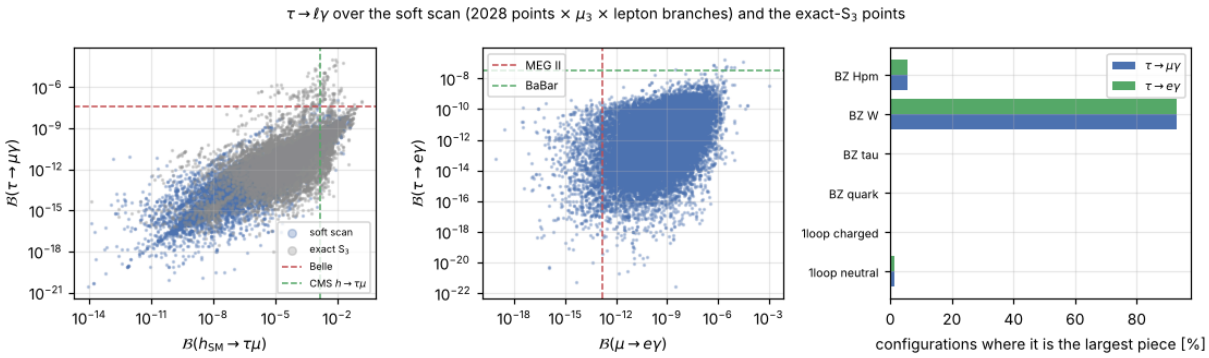


Figure 1: Left: $\mathcal{B}(\tau \to \mu\gamma)$ against the SM-like $\mathcal{B}(h \to \tau\mu)$ per configuration, soft scan (blue, 40 000 sampled) and exact-S_3 points (grey), with the Belle and CMS limits dashed. Middle: $\mathcal{B}(\tau \to e\gamma)$ against $\mathcal{B}(\mu \to e\gamma)$ in the soft scan, with the BaBar and MEG II limits. Right: which piece of the amplitude is largest, by share of configurations.

In each variant the counts equal the MEG II-only ones of the previous report: the τ modes remove no point under any of them.

6 A prediction for Belle II

$\mathcal{B}(\tau \to e\gamma)/\mathcal{B}(\mu \to e\gamma)$ has median 2.8×10^{-3}. In the MEG-allowed configurations the τ rates stay at or below 7.5×10^{-10}. Where every radiative and Higgs LFV limit holds, the largest values are

$$\mathcal{B}(\tau \to \mu\gamma) = 4.8 \times 10^{-10}, \qquad \mathcal{B}(\tau \to e\gamma) = 3.9 \times 10^{-10},$$

about 15× and 23× below the expected Belle II limits at 50 ab^{-1}, 6.9×10^{-9} and 9.0×10^{-9} [3]. With this model's soft vacuum and the quark ansatz, **a $\tau \to \ell\gamma$ signal at Belle II would exclude it**. The exception is the light-scalar corner of the exact vacuum (§4).

7 Open questions

1. **The missing two-loop pieces and the quark ansatz** of the $\mu \to e\gamma$ report carry over unchanged. The soft-vacuum conclusion (no new point removed) has a margin of about 10^2, so they cannot reverse it.

2. **The light-scalar corner.** The exact-vacuum $\tau \to \mu\gamma$ exclusions sit at non-SM neutral even scalars below about 220 GeV. Their status in LEP and LHC direct searches (the scans cut only on the hVV coupling) would say whether this corner is already closed.

3. $\tau \to 3\mu$ **and** $\tau \to \mu ee$ share the $\tau\mu$ dipole and add tree-level contact terms through the same scalars.

References

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[9] Particle Data Group, *Review of Particle Physics*, pdg.lbl.gov: $\mathcal{B}(\tau \to e\nu\bar{\nu})$, $\mathcal{B}(\tau \to \mu\nu\bar{\nu})$ and τ_τ.